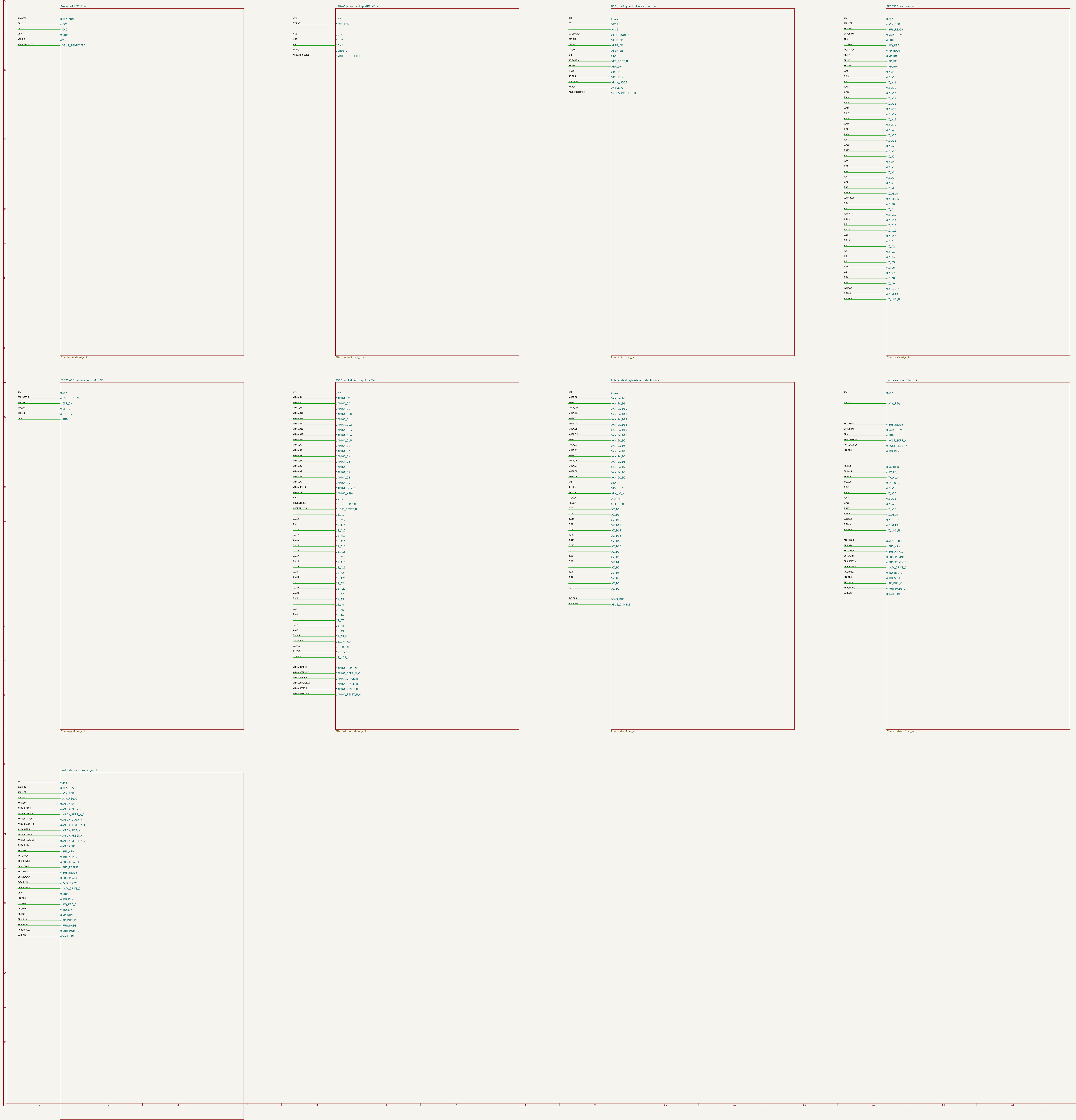
FujiNet A500 integrated R4 — circuit review; NOT fabrication release

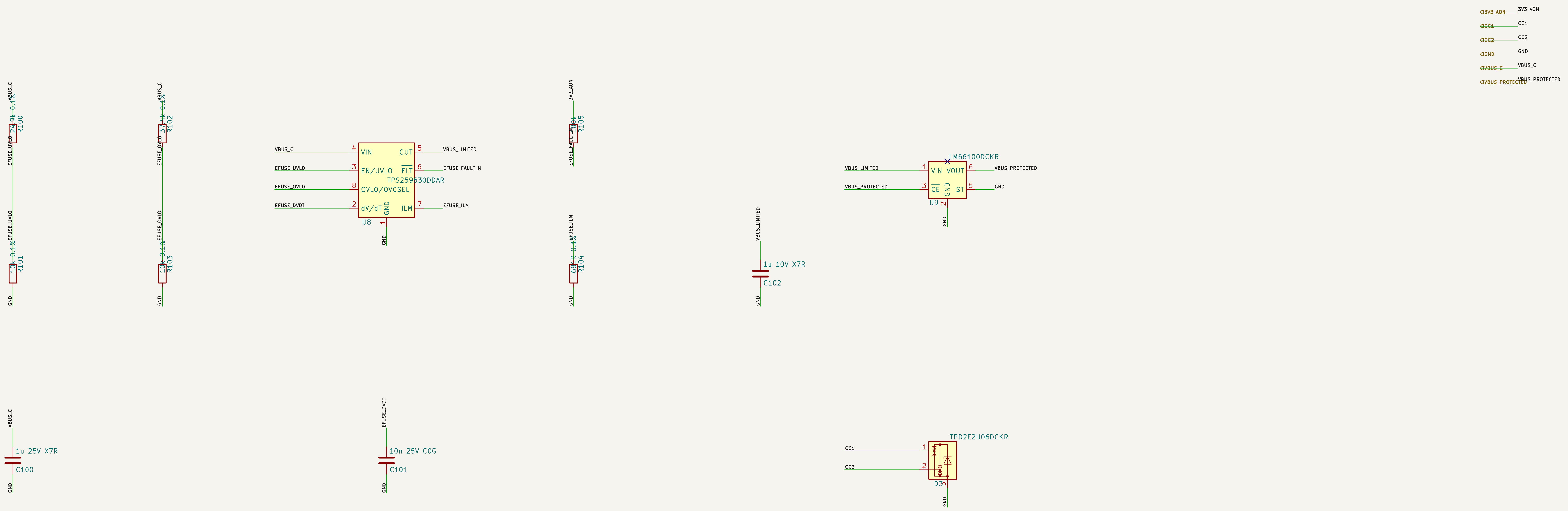


Sheet: /
File: integrated–a500–r4.kicad_sch

Title: FujiNet direct A500 integrated candidate

Size: A0
Date: 2026–11–09
Rev: R4 candidate
KiCad E.D.A. 10.0.7
Id: 1/10

USB VBUS -> current/voltage protection -> reverse-blocking ideal diode -> AON and qualified main supply.



681R: nominal 1.337A limit. OVLO nominal 5.688V. Latch-off thermal fault: unplug USB to recover.

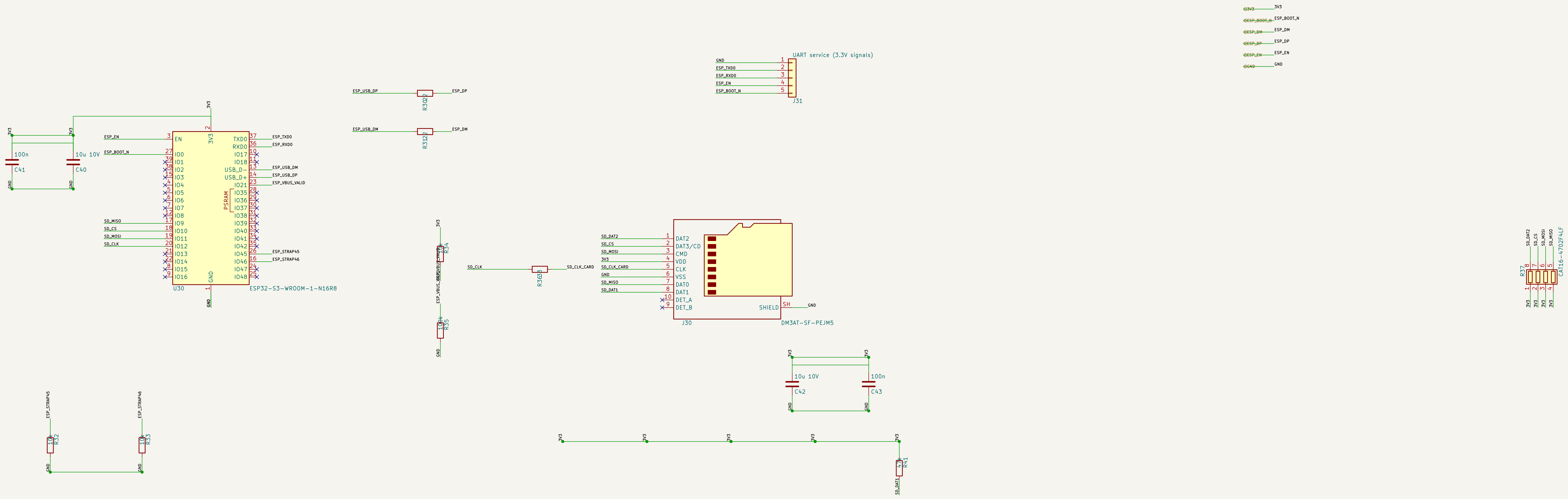
No Amiga supply connection. E-fuse pad/package and fast overvoltage transient review required before layout.

FujiNet NIO
Sheet: /Protected USB input/
File: input.kicad_sch

Title: Protected USB input

Size: A2 Date: 2026-11-09 **Rev: R4 candidate**
KiCad E.D.A. 10.0.7 Id: 2/10

N16R8 module includes flash, PSRAM, crystal and RF. IO35/36/37 are reserved for octal PSRAM.



Antenna at enclosure/PCB edge: apply Espressif all-layer antenna keepout before PCB placement.

FujiNet NIO
Sheet: /ESP32-S3 module and microSD/
File: esp.kicad_sch

Title: ESP32-S3 module and microSD

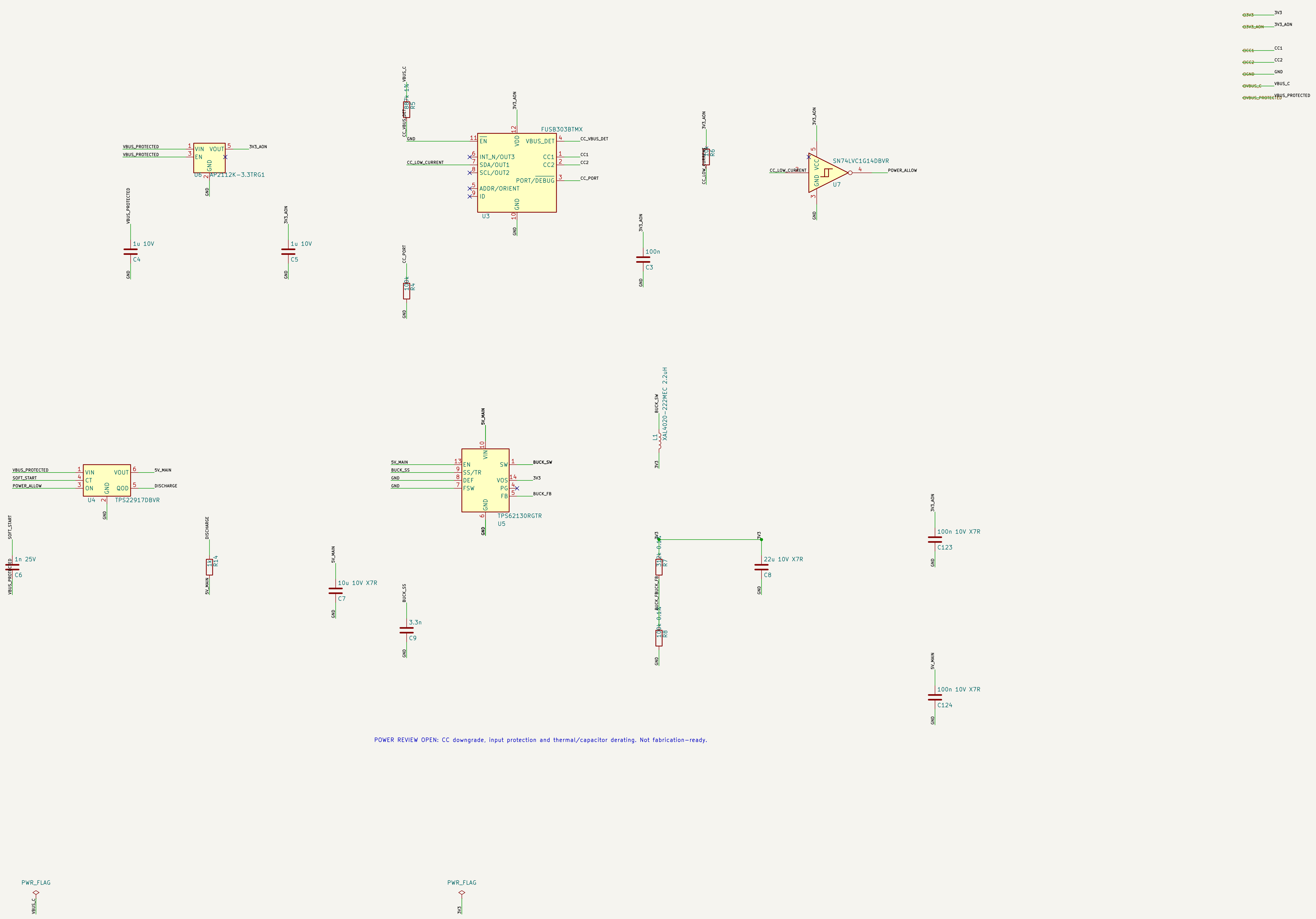
Size: A1	Date: 2026-11-09	**Rev: R4 candidate**
KiCad E.D.A. 10.0.7		Id: 3/10

Sheet: /Host interface power guard/
File: guard.kicad_sch

Title: guard

Size: A1 | Date: | **Rev:**
KiCad E.D.A. 10.0.7 | Id: 4/10

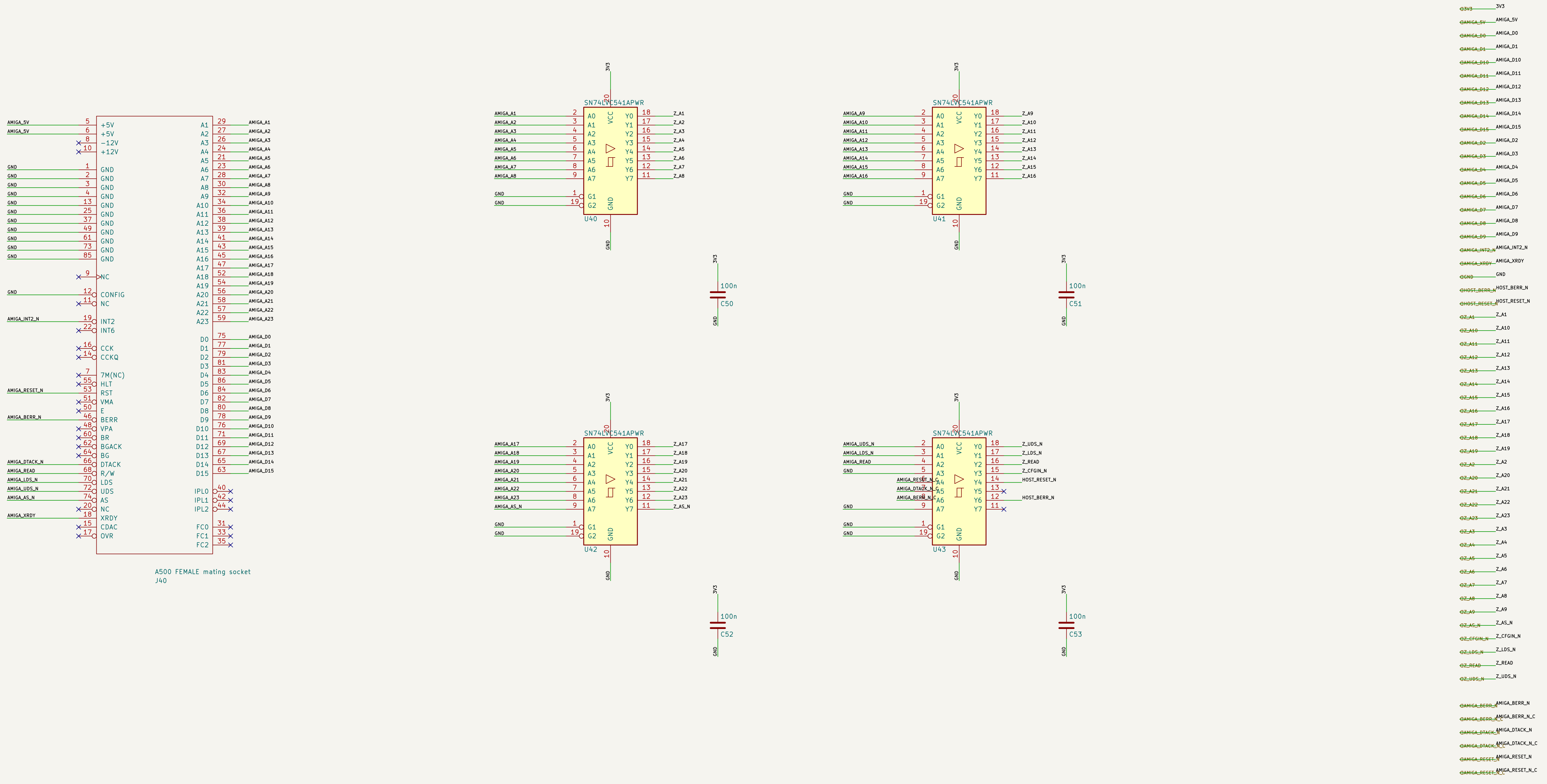
USB−C sink only. >=1.5 A advertised required. No Amiga supply connection to these rails.



POWER REVIEW OPEN: CC downgrade, input protection and thermal/capacitor derating. Not fabrication−ready.

FujiNet NIO
Sheet: /USB−C power and qualification/
File: power.kicad_sch
Title: USB−C power and qualification
Size: A1 Date: 2026−11−09 **Rev: R4 candidate**
KiCad E.D.A. 10.0.7 Id: 5/10

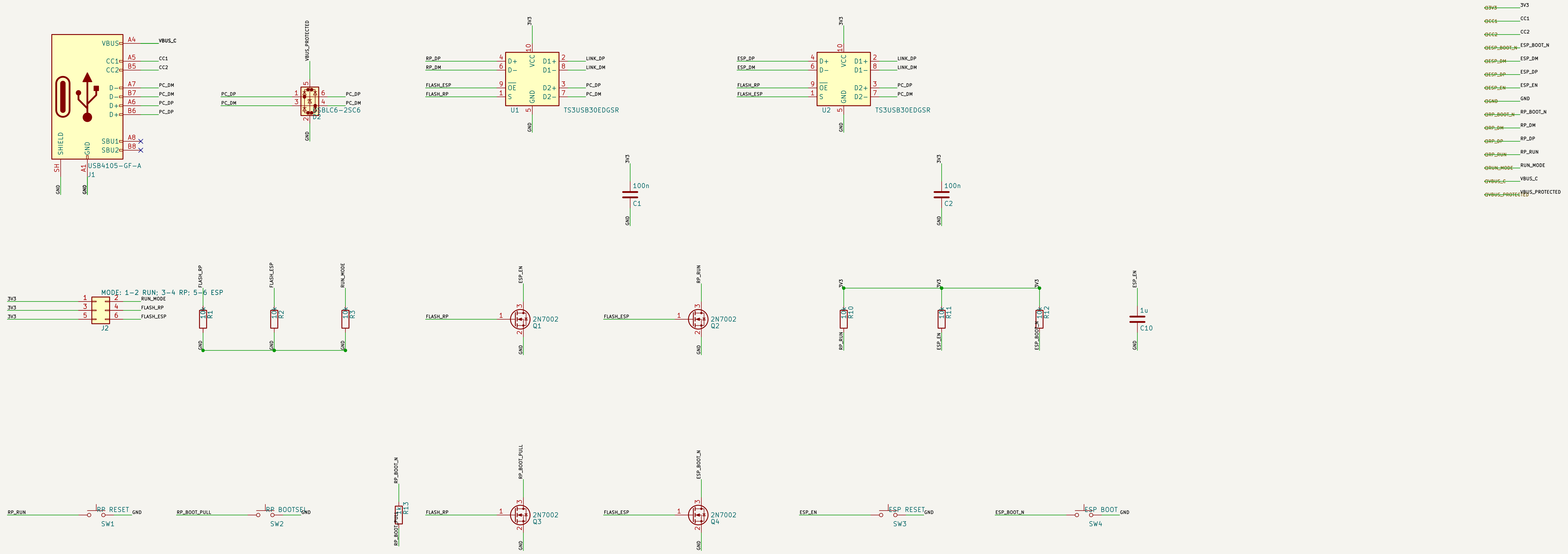
86–contact A500 socket; NOT the existing male PCB–edge footprint. No A2000 SLAVE/CFGOUT here.



U40–U43 use pin–compatible 541 logic symbol; fitted part MUST be 3.3V SN74LVC541A (5V tolerant, Ioff).

FujiNet NIO
Sheet: /A500 socket and input buffers/
File: address.kicad_sch
Title: A500 socket and input buffers
Size: A1
Date: 2026–11–09
Rev: R4 candidate
KiCad E.D.A. 10.0.7
Id: 6/10

Shared USB: cold mode changes only. Unplug power before moving the single MODE shunt.



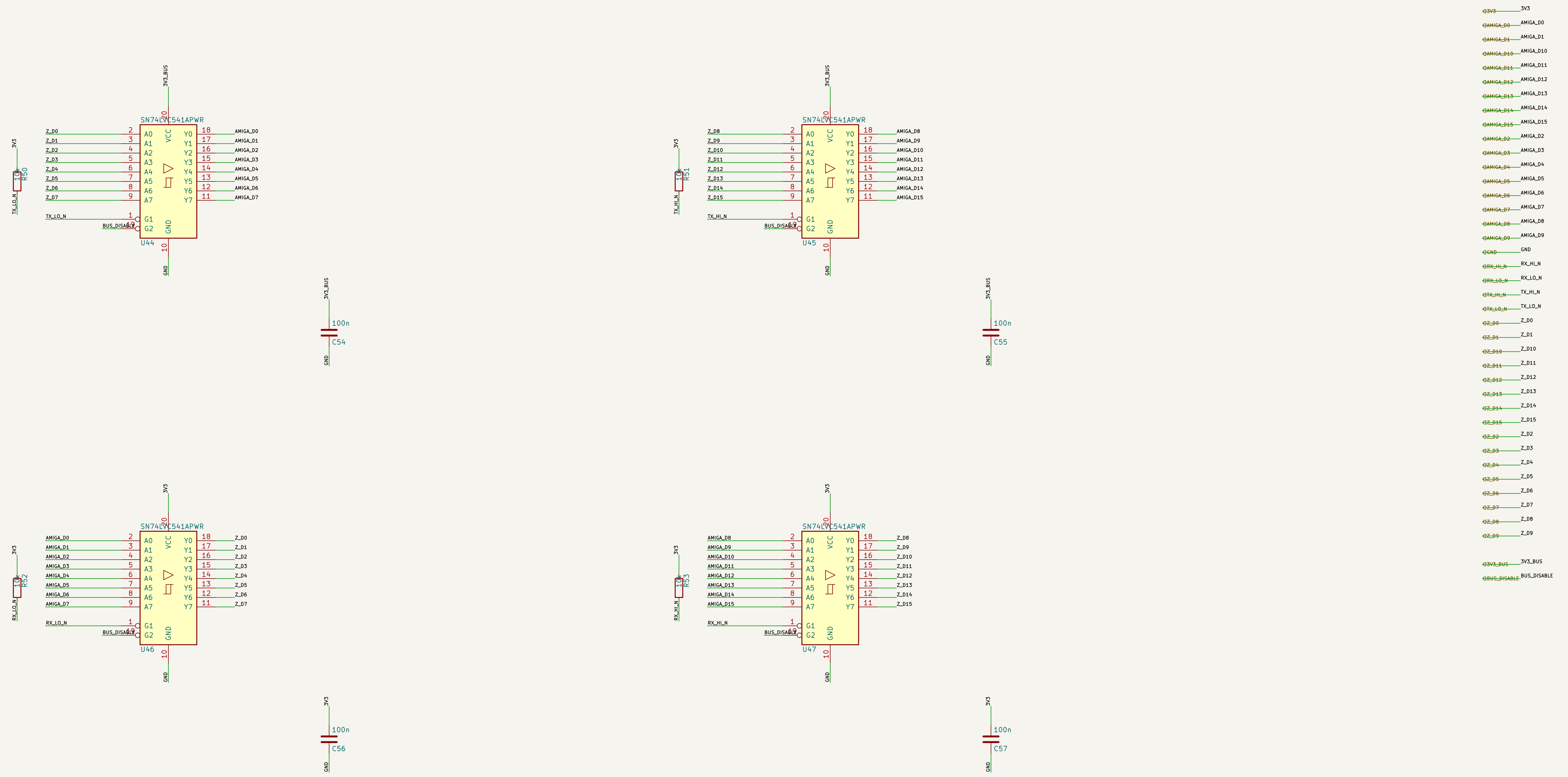
Physical recovery: selected BOOT held low, peer held RESET. SW1/SW3 are independent resets.

FujiNet NIO
Sheet: /USB routing and physical recovery/
File: usb.kicad_sch

Title: USB routing and physical recovery

Size: A2 Date: 2026-11-09 **Rev: R4 candidate**
KiCad E.D.A. 10.0.7 Id: 7/10

Upper row: RP -> Amiga. Lower row: Amiga -> RP. Byte enables independent; hardware read/write qualification.



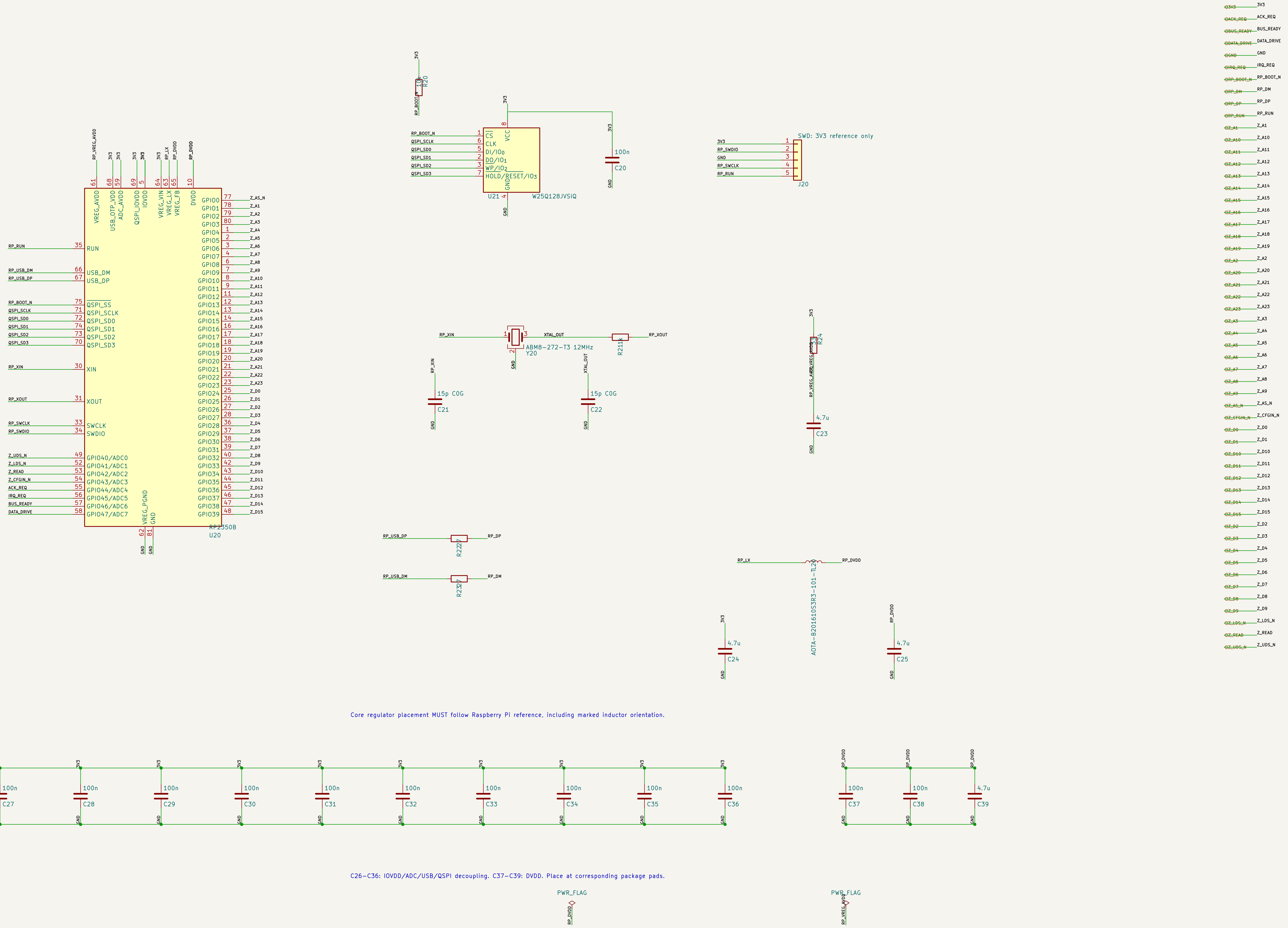
PIO: preload read data/direction, pulse DATA_DRIVE, settle, pulse ACK_REQ. Return GPIO to input between cycles.

FujiNet NIO
Sheet: /Independent byte-lane data buffers/
File: data.kicad_sch

Title: Independent byte-lane data buffers

Size: A1
Date: 2026-11-09
Rev: R4 candidate
KiCad E.D.A. 10.0.7
Id: 8/10

48 GPIO: address/data/control only; USB and SWD dedicated pads. GPIO45 IRQ; GPIO46 BUS_READY.



Core regulator placement MUST follow Raspberry Pi reference, including marked inductor orientation.

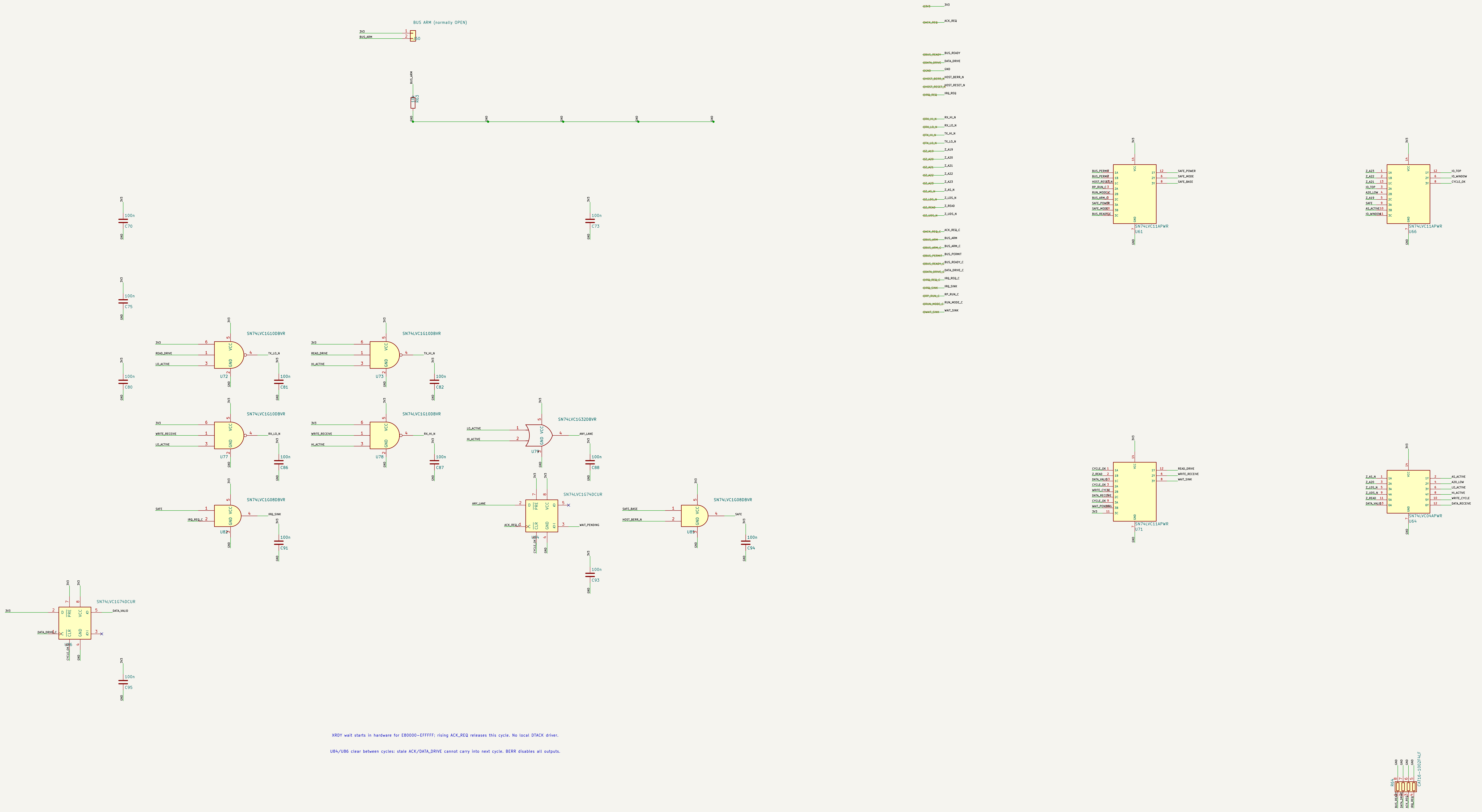
C26–C36: IOVDD/ADC/USB/QSPI decoupling. C37–C39: DVDD. Place at corresponding package pads.

FujiNet NIO
Sheet: /RP2350B and support/
File: rp.kicad_sch

Title: RP2350B and support

Size: A2
Date: 2026-11-09
Rev: R4 candidate
KiCad E.D.A. 10.0.7
Id: 9/10

Host monitor: 0.405V x 11 = 4.455V nominal trip; CT open = 20ms release delay. No host supply drawn for operation.



XRDY wait starts in hardware for E80000–EFFFFF; rising ACK_REQ releases this cycle. No local DTACK driver.

U84/U86 clear between cycles: stale ACK/DATA_DRIVE cannot carry into next cycle. BERR disables all outputs.

FujiNet NIO
Sheet: /Hardware bus interlocks/
File: control.kicad_sch
Title: Hardware bus interlocks
Size: A1
Date: 2026–11–09
Rev: R4 candidate
KiCad E.D.A. 10.0.7
Id: 10/10